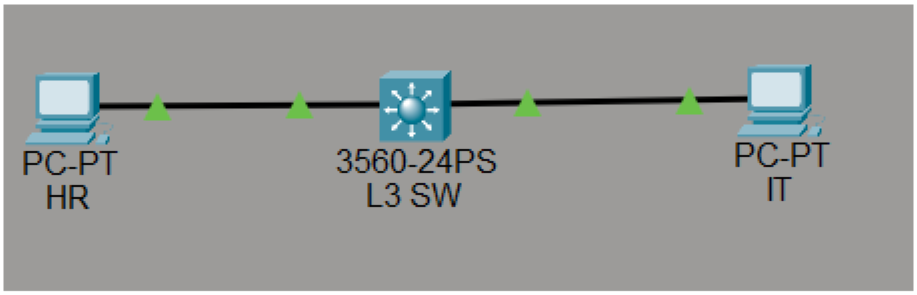
MINI PROJECT 3

Inter-VLAN Routing using Layer-3 Switch (SVI) + VLAN + Hardware Routing

DIAGRAM:



DESCRIPTION:

This project replaces the router-on-a-stick design with a Layer-3 switch to perform inter-VLAN routing using SVIs. L-3 Switch maintains Mac-address & Routing Table both routing in hardware.

STEP 1: Create VLANs on SW 1:

Verify existing VLANs from Previous projects

STEP 2: Assign Access Ports to VLANs:

```
interface fastEthernet0/1
switchport mode access
switchport access vlan 10
exit

interface fastEthernet0/2
switchport mode access
switchport access vlan 20
exit
```

STEP 3: Create SVIs (Switched Virtual Interfaces)

```
interface vlan 10
 ip address 192.168.10.1 255.255.255.0
 no shutdown
exit

interface vlan 20
 ip address 192.168.20.1 255.255.255.0
 no shutdown
exit
```

STEP 4: Configure IP Addressing on PCs:

PC	VLAN	IP Address	Gateway
PC-HR	10	192.168.10.10/24	192.168.10.1
PC-IT	20	192.168.20.10/24	192.168.20.1

STEP 5: Enable IP Routing on the Switch:

```
ip routing
```

STEP 6: Verification & Ping Test:

```
show ip route
show ip interface brief
```